

Start Up

This chapter describes how to start up the PowerFlex 4 Drive. To simplify drive setup, the most commonly programmed parameters are organized in a single Basic Program Group.

Important: Read the *General Precautions* section before proceeding.



ATTENTION: Power must be applied to the drive to perform the following start-up procedures. Some of the voltages present are at incoming line potential. To avoid electric shock hazard or damage to equipment, only qualified service personnel should perform the following procedure. Thoroughly read and understand the procedure before beginning. If an event does not occur while performing this procedure, **Do Not Proceed. Remove All Power** including user supplied control voltages. User supplied voltages may exist even when main AC power is not applied to the drive. Correct the malfunction before continuing.

Prepare For Drive Start-Up

Before Applying Power to the Drive

- ☐ 1. Confirm that all inputs are connected to the correct terminals and are secure.
- ☐ 2. Verify that AC line power at the disconnect device is within the rated value of the drive.
- ☐ 3. Verify that any digital control power is 24 volts.
- ☐ 4. Verify that the Sink (SNK)/Source (SRC) Setup DIP Switch is set to match your control wiring scheme. See [Figure 1.5 on page 1-13](#) for location.

Important: The default control scheme is Source (SRC). The Stop terminal is jumpered (I/O Terminals 01 and 11) to allow starting from the keypad. If the control scheme is changed to Sink (SNK), the jumper must be removed from I/O Terminals 01 and 11 and installed between I/O Terminals 01 and 04.

- ☐ 5. Verify that the Stop input is present or the drive will not start.

Important: If I/O Terminal 01 is used as a stop input, the jumper between I/O Terminals 01 and 11 must be removed.

Applying Power to the Drive

- ❑ 6. Apply AC power and control voltages to the drive.
- ❑ 7. Familiarize yourself with the integral keypad features (see [page 2-3](#)) before setting any Program Group parameters.

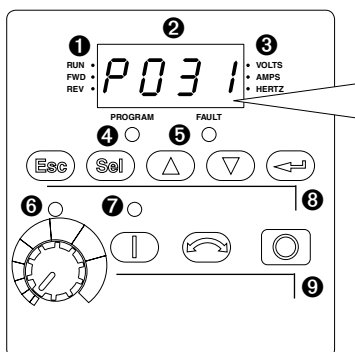
Start, Stop, Direction and Speed Control

Factory default parameter values allow the drive to be controlled from the integral keypad. No programming is required to start, stop, change direction and control speed directly from the integral keypad.

Important: To disable reverse operation, see [A095](#) [Reverse Disable].

If a fault appears on power up, refer to [Fault Descriptions on page 4-3](#) for an explanation of the fault code.

Integral Keypad



Menu	Description
<i>d</i>	Display Group (View Only) Consists of commonly viewed drive operating conditions.
<i>P</i>	Basic Program Group Consists of most commonly used programmable functions.
<i>A</i>	Advanced Program Group Consists of remaining programmable functions.
<i>F</i>	Fault Designator Consists of list of codes for specific fault conditions. Displayed only when fault is present.

No.	LED	LED State	Description
1	Run/Direction Status	Steady Red	Indicates drive is running and commanded motor direction.
		Flashing Red	Drive has been commanded to change direction. Indicates actual motor direction while decelerating to zero.
2	Alphanumeric Display	Steady Red	Indicates parameter number, parameter value, or fault code.
		Flashing Red	Single digit flashing indicates that digit can be edited. All digits flashing indicates a fault condition.
3	Displayed Units	Steady Red	Indicates the units of the parameter value being displayed.
4	Program Status	Steady Red	Indicates parameter value can be changed.
5	Fault Status	Flashing Red	Indicates drive is faulted.
6	Pot Status	Steady Green	Indicates potentiometer on Integral Keypad is active.
7	Start Key Status	Steady Green	Indicates Start key on Integral Keypad is active. The Reverse key is also active unless disabled by A095 [Reverse Disable].

No.	Key	Name	Description
8		Escape	Back one step in programming menu. Cancel a change to a parameter value and exit Program Mode.
		Select	Advance one step in programming menu. Select a digit when viewing parameter value.
		Up Arrow	Scroll through groups and parameters.
		Down Arrow	Increase/decrease the value of a flashing digit.
9		Enter	Advance one step in programming menu. Save a change to a parameter value.
		Potentiometer	Used to control speed of drive. Default is active. Controlled by parameter P038 [Speed Reference].
		Start	Used to start the drive. Default is active. Controlled by parameter P036 [Start Source].
		Reverse	Used to reverse direction of the drive. Default is active. Controlled by parameters P036 [Start Source] and A095 [Reverse Disable].
		Stop	Used to stop the drive or clear a fault. This key is always active. Controlled by parameter P037 [Stop Mode].

Viewing and Editing Parameters

The last user-selected Display Group parameter is saved when power is removed and is displayed by default when power is reapplied.

The following is an example of basic integral keypad and display functions. This example provides basic navigation instructions and illustrates how to program the first Program Group parameter.

Step	Key(s)	Example Displays
1. When power is applied, the last user-selected Display Group parameter number is briefly displayed with flashing characters. The display then defaults to that parameter's current value. (Example shows the value of d001 [Output Freq] with the drive stopped.)		
2. Press Esc once to display the Display Group parameter number shown on power-up. The parameter number will flash.	Esc	
3. Press Esc again to enter the group menu. The group menu letter will flash.	Esc	
4. Press the Up Arrow or Down Arrow to scroll through the group menu (d, P and A).	△ or ▽	
5. Press Enter or Sel to enter a group. The right digit of the last viewed parameter in that group will flash.	↵ or Sel	
6. Press the Up Arrow or Down Arrow to scroll through the parameters that are in the group.	△ or ▽	
7. Press Enter or Sel to view the value of a parameter. If you do not want to edit the value, press Esc to return to the parameter number.	↵ or Sel	
8. Press Enter or Sel to enter program mode to edit the parameter value. The right digit will flash and the Program LED will illuminate if the parameter can be edited.	↵ or Sel	
9. Press the Up Arrow or Down Arrow to change the parameter value. If desired, press Sel to move from digit to digit or bit to bit. The digit or bit that you can change will flash.	△ or ▽	
10. Press Esc to cancel a change. The digit will stop flashing, the previous value is restored and the Program LED will turn off. Or Press Enter to save a change. The digit will stop flashing and the Program LED will turn off.	Esc ↵	
11. Press Esc to return to the parameter list. Continue to press Esc to back out of the programming menu. If pressing Esc does not change the display, then d001 [Output Frequency] is displayed. Press Enter or Sel to enter the group menu.	Esc	

The Basic Program Group ([page 3-8](#)) contains the most commonly changed parameters.

Programming and Parameters

Chapter 3 provides a complete listing and description of the PowerFlex 4 parameters. Parameters are programmed (viewed/edited) using the integral keypad. As an alternative, programming can also be performed using DriveExplorer™ or DriveExecutive™ software, a personal computer and a serial converter module. Refer to [Appendix B](#) for catalog numbers.

For information on...	See page...
About Parameters	3-1
Parameter Organization	3-2
Basic Program Group	3-8
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About Parameters

To configure a drive to operate in a specific way, drive parameters may have to be set. Three types of parameters exist:

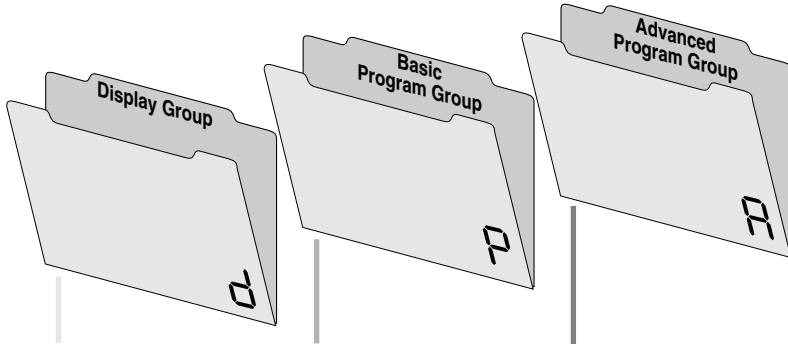
- **ENUM**
ENUM parameters allow a selection from 2 or more items. Each item is represented by a number.
- **Numeric Parameters**
These parameters have a single numerical value (i.e. 0.1 Volts).
- **Bit Parameters**
Bit parameters have four individual bits associated with features or conditions. If the bit is 0, the feature is off or the condition is false. If the bit is 1, the feature is on or the condition is true.

Some parameters are marked as follows.

 = Stop drive before changing this parameter.

 = 32 bit parameter. Parameters marked 32 bit will have two parameter numbers when using RS485 communications and programming software.

Parameter Organization



See page 3-3

Output Freq	d001
Commanded Freq	d002
Output Current	d003
Output Voltage	d004
DC Bus Voltage	d005
Drive Status	d006
Fault 1 Code	d007
Fault 2 Code	d008
Fault 3 Code	d009
Process Display	d010
Control Source	d012
Ctrl In Status	d013
Dig In Status	d014
Comm Status	d015
Control SW Ver	d016
Drive Type	d017
Elapsed Run Time	d018
Testpoint Data	d019
Analog In 0-10V	d020
Analog In 4-20mA	d021
Drive Temp	d024

See page 3-8

Motor NP Volts	P031
Motor NP Hertz	P032
Motor OL Current	P033
Minimum Freq	P034
Maximum Freq	P035
Start Source	P036
Stop Mode	P037
Speed Reference	P038
Accel Time 1	P039
Decel Time 1	P040
Reset To Defaults	P041

See page 3-13

Digital In1 Sel	A051
Digital In2 Sel	A052
Relay Out Sel	A055
Relay Out Level	A056
Accel Time 2	A067
Decel Time 2	A068
Internal Freq	A069
Preset Freq 0	A070
Preset Freq 1	A071
Preset Freq 2	A072
Preset Freq 3	A073
Jog Frequency	A078
Jog Accel/Decel	A079
DC Brake Time	A080
DC Brake Level	A081
DB Resistor Sel	A082
S Curve %	A083
Boost Select	A084
Maximum Voltage	A088
Current Limit	A089
Motor OL Select	A090
PWM Frequency	A091
Auto Rstrt Tries	A092
Auto Rstrt Delay	A093
Start At PowerUp	A094
Reverse Disable	A095
Flying Start En	A096
Compensation	A097
SW Current Trip	A098
Process Factor	A099
Fault Clear	A100
Program Lock	A101
Testpoint Sel	A102
Comm Data Rate	A103
Comm Node Addr	A104
Comm Loss Action	A105
Comm Loss Time	A106
Comm Format	A107
Anlg In 0-10V Lo	A110
Anlg In 0-10V Hi	A111
Anlg In 4-20mA Lo	A112
Anlg In 4-20mA Hi	A113
Slip Hertz @ FLA	A114
Process Time Lo	A115
Process Time Hi	A116

Display Group

d001 [Output Freq]

Related Parameter(s): [d002](#), [d010](#), [P034](#), [P035](#), [P038](#)

Output frequency present at T1, T2 & T3 (U, V & W).

Values	Default	Read Only
	Min/Max:	0.0/ P035 [Maximum Freq]
	Display:	0.1 Hz

d002 [Commanded Freq]

Related Parameter(s): [d001](#), [d013](#), [P034](#), [P035](#), [P038](#)

Value of the active frequency command. Displays the commanded frequency even if the drive is not running.

Important: The frequency command can come from a number of sources. Refer to [Start and Speed Reference Control on page 1-17](#) for details.

Values	Default	Read Only
	Min/Max:	0.0/ P035 [Maximum Freq]
	Display:	0.1 Hz

d003 [Output Current]

The output current present at T1, T2 & T3 (U, V & W).

Values	Default	Read Only
	Min/Max:	0.00/(Drive Rated Amps × 2)
	Display:	0.01 Amps

d004 [Output Voltage]

Related Parameter(s): [P031](#), [A084](#), [A088](#)

Output voltage present at terminals T1, T2 & T3 (U, V & W).

Values	Default	Read Only
	Min/Max:	0/Drive Rated Volts
	Display:	1 VAC

d005 [DC Bus Voltage]

Present DC bus voltage level.

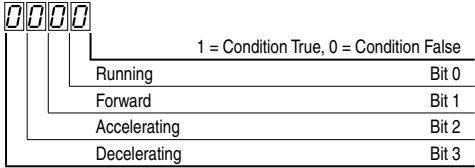
Values	Default	Read Only
	Min/Max:	Based on Drive Rating
	Display:	1 VDC

Display Group *(continued)*

d006 [Drive Status]

Related Parameter(s): [A095](#)

Present operating condition of the drive.



Values	Default	Read Only
Min/Max:	0/1	
Display:	1	

d007 [Fault 1 Code]

d008 [Fault 2 Code]

d009 [Fault 3 Code]

A code that represents a drive fault. The codes will appear in these parameters in the order they occur ([d007](#) [Fault 1 Code] = the most recent fault). Repetitive faults will only be recorded once.

Refer to [Chapter 4](#) for fault code descriptions.

Values	Default	Read Only
Min/Max:	F2/F122	
Display:	F1	

d010 [Process Display]

Related Parameter(s): [d001](#), [A099](#), [A115](#), [A116](#)

 32 bit parameter.

The output frequency scaled by [A099](#) [Process Factor] or by [A115](#) [Process Time Lo] and [A116](#) [Process Time Hi].

$$\text{Output Freq} \times \text{Process Factor} = \text{Process Display}$$

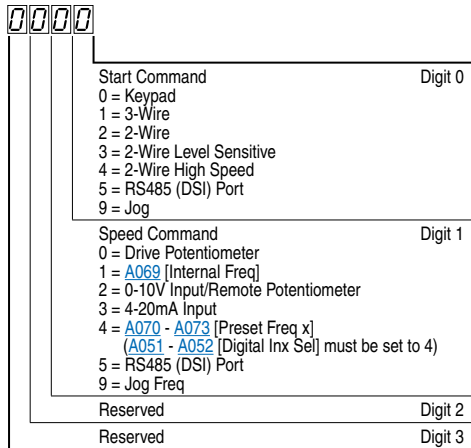
Values	Default	Read Only
Min/Max:	0.00/9999	
Display:	0.01 – 1	

Display Group *(continued)*

d012 [Control Source]

Related Parameter(s): [P036](#), [P038](#), [A051](#), [A052](#)

Displays the active source of the Start Command and Speed Command which are normally defined by the settings of [P036](#) [Start Source] and [P038](#) [Speed Reference] but may be overridden by digital inputs. Refer to the flowcharts on pages [1-17](#) and [1-18](#) for details.



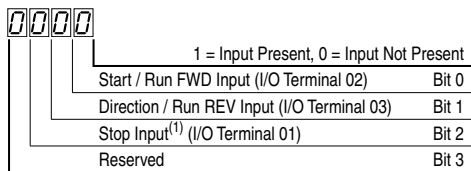
Values	Default	Read Only
Min/Max:		0/9
Display:		1

d013 [Contrl In Status]

Related Parameter(s): [d002](#), [P034](#), [P035](#)

Status of the control terminal block control inputs.

Important: Actual control commands may come from a source other than the control terminal block.



⁽¹⁾ The stop input must be present in order to start the drive.
When this bit is a 1 the drive can be started.
When this bit is a 0 the drive will stop.

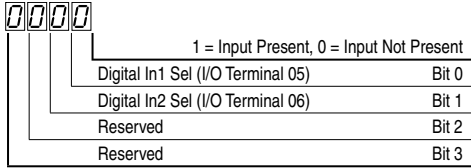
Values	Default	Read Only
Min/Max:		0/1
Display:		1

Display Group *(continued)*

d014 [Dig In Status]

Related Parameter(s): [A051](#), [A052](#)

Status of the control terminal block digital inputs.

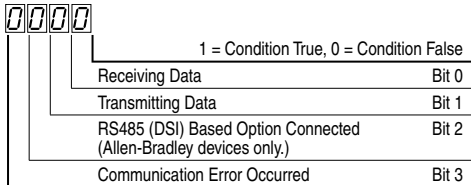


Values	Default	Read Only
Min/Max:	0/1	
Display:	1	

d015 [Comm Status]

Related Parameter(s): [A103](#) - [A107](#)

Status of the communications ports.



Values	Default	Read Only
Min/Max:	0/1	
Display:	1	

d016 [Control SW Ver]

Main Control Board software version.

Values	Default	Read Only
Min/Max:	1.00/99.99	
Display:	0.01	

d017 [Drive Type]

Used by Rockwell Automation field service personnel.

Values	Default	Read Only
Min/Max:	1001/9999	
Display:	1	

Display Group *(continued)*

d018 [Elapsed Run Time]

Accumulated time drive is outputting power. Time is displayed in 10 hour increments.

Values	Default	Read Only
	Min/Max:	0/9999 Hrs
	Display:	1 = 10 Hrs

d019 [Testpoint Data]

Related Parameter(s): [A102](#)

The present value of the function selected in [A102](#) [Testpoint Sel].

Values	Default	Read Only
	Min/Max:	0/FFFF
	Display:	1 Hex

d020 [Analog In 0-10V]

Related Parameter(s): [A110](#), [A111](#)

The present value of the voltage at I/O Terminal 13 (100.0% = 10 volts).

Values	Default	Read Only
	Min/Max:	0.0/100.0%
	Display:	0.1%

d021 [Analog In 4-20mA]

Related Parameter(s): [A112](#), [A113](#)

The present value of the current at I/O Terminal 15 (0.0% = 4mA, 100.0% = 20mA).

Values	Default	Read Only
	Min/Max:	0.0/100.0%
	Display:	0.1%

d024 [Drive Temp]

Present operating temperature of the drive power section.

Values	Default	Read Only
	Min/Max:	0/120 degC
	Display:	1 degC

Basic Program Group

P031 [Motor NP Volts]

Related Parameter(s): [d004](#), [A084](#)



Stop drive before changing this parameter.

Set to the motor nameplate rated volts.

Values	Default	Based on Drive Rating
	Min/Max:	20/Drive Rated Volts
	Display:	1 VAC

P032 [Motor NP Hertz]

Related Parameter(s): [A084](#), [A090](#)



Stop drive before changing this parameter.

Set to the motor nameplate rated frequency.

Values	Default	60 Hz
	Min/Max:	10/240 Hz
	Display:	1 Hz

P033 [Motor OL Current]

Related Parameter(s): [A055](#), [A089](#), [A090](#), [A098](#), [A114](#)

Set to the maximum allowable motor current.

The drive will fault on an F7 Motor Overload if the value of this parameter is exceeded by 150% for 60 seconds or 200% for 3 seconds.

Values	Default	Based on Drive Rating
	Min/Max:	0.0/(Drive Rated Amps × 2)
	Display:	0.1 Amps

P034 [Minimum Freq]

Related Parameter(s): [d001](#), [d002](#), [d013](#), [P035](#), [A110](#), [A112](#), [A115](#)

Sets the lowest frequency the drive will output continuously.

Values	Default	0.0 Hz
	Min/Max:	0.0/240.0 Hz
	Display:	0.1 Hz

P035 [Maximum Freq]

Related Parameter(s): [d001](#), [d002](#), [d013](#), [P034](#), [A078](#), [A111](#), [A113](#), [A115](#)



Stop drive before changing this parameter.

Sets the highest frequency the drive will output.

Values	Default	60 Hz
	Min/Max:	0/240 Hz
	Display:	1 Hz

Basic Program Group *(continued)*

P036 [Start Source]

Related Parameter(s): [d012](#), [P037](#)



Stop drive before changing this parameter.

Sets the control scheme used to start the drive.

Refer to [Start and Speed Reference Control on page 1-17](#) for details about how other drive settings can override the setting of this parameter.

Important: For all settings except option 3, the drive must receive a leading edge from the start input for the drive to start after a stop input, loss of power or fault condition.

Options	0	"Keypad" (Default)	- Integral keypad controls drive operation. - I/O Terminal 1 "Stop" = coast to stop. - When active, the Reverse key is also active unless disabled by A095 [Reverse Disable].
	1	"3-Wire"	I/O Terminal 1 "Stop" = stop according to the value set in P037 [Stop Mode].
	2	"2-Wire"	I/O Terminal 1 "Stop" = coast to stop.
	3	"2-W Lvl Sens"	Drive will restart after a "Stop" command when: - Stop is removed and - Start is held active



ATTENTION: Hazard of injury exists due to unintended operation. When P036 [Start Source] is set to option 3, and the Run input is maintained, the Run inputs do not need to be toggled after a Stop input for the drive to run again. A Stop function is provided only when the Stop input is active (open).

	4	"2-W Hi Speed"	Important: There is greater potential voltage on the output terminals when using this option. - Outputs are kept in a ready-to-run state. The drive will respond to a "Start" command within 10 ms. - I/O Terminal 1 "Stop" = coast to stop.
	5	"Comm Port"	- Remote communications. Refer to Appendix C for details. - I/O Terminal 1 "Stop" = coast to stop.

Important: When commanding Jog via the RS485 communications port on drives with firmware version 1.02 or earlier, the Jog command will follow the commanded direction from I/O Terminal 03. On firmware versions 1.03 and later, the commanded direction will be provided via the RS485 communications port.

Important: When sending a continuous start command via the RS485 communications port on drives with firmware version 1.02 or earlier, a maintained stop input is required to stop the drive. Once the stop input is inactive, the drive will restart. On firmware versions 1.03 and later, once a stop input is received, the start command must transition from high to low to high for the drive to start.

Basic Program Group *(continued)*

P037 [Stop Mode]

Related Parameter(s): [P036](#), [A080](#), [A081](#), [A082](#), [A105](#)

Active stop mode for all stop sources [e.g. keypad, run forward (I/O Terminal 02), run reverse (I/O Terminal 03), RS485 port] except as noted below.

Important: I/O Terminal 01 is always a coast to stop input except when P036 [Start Source] is set for “3-Wire” control. When in three wire control, I/O Terminal 01 is controlled by P037 [Stop Mode].

Hardware Enable Circuitry

By default, I/O Terminal 01 is a coast to stop input. The status of the input is interpreted by drive software. If the application requires the drive to be disabled without software interpretation, a “dedicated” hardware enable configuration can be utilized. This is accomplished by removing the ENBL enable jumper on the control board. In this case, the drive will always coast to a stop regardless of the settings of P036 [Start Source] and P037 [Stop Mode].

Options	0	“Ramp, CF” ⁽¹⁾ (Default)	Ramp to Stop. “Stop” command clears active fault.
	1	“Coast, CF” ⁽¹⁾	Coast to Stop. “Stop” command clears active fault.
	2	“DC Brake, CF” ⁽¹⁾	DC Injection Braking Stop. “Stop” command clears active fault.
	3	“DCBrkAuto,CF” ⁽¹⁾	DC Injection Braking Stop with Auto Shutoff. • Standard DC Injection Braking for value set in A080 [DC Brake Time]. - OR • Drive shuts off if the drive detects that the motor is stopped. “Stop” command clears active fault.
	4	“Ramp”	Ramp to Stop.
	5	“Coast”	Coast to Stop.
	6	“DC Brake”	DC Injection Braking Stop.
	7	“DC BrakeAuto”	DC Injection Braking Stop with Auto Shutoff. • Standard DC Injection Braking for value set in A080 [DC Brake Time]. - OR • Drive shuts off if current limit is exceeded.

⁽¹⁾ Stop input also clears active fault.

Basic Program Group *(continued)*

P038 [Speed Reference] Related Parameter(s): [d001](#), [d002](#), [d012](#), [P039](#), [P040](#), [A051](#), [A052](#), [A069](#), [A070-A073](#), [A110](#), [A111](#), [A112](#), [A113](#)

Sets the source of the speed reference to the drive.

The drive speed command can be obtained from a number of different sources. The source is normally determined by [P038](#) [Speed Reference]. However, when [A051](#) - [A052](#) [Digital Inx Sel] is set to option 2, 4, 5, 6, 11, 12, 13, 14, 15 and the digital input is active, the speed reference commanded by [P038](#) [Speed Reference] will be overridden. Refer to the flowchart on [page 1-17](#) for more information on speed reference control priority.

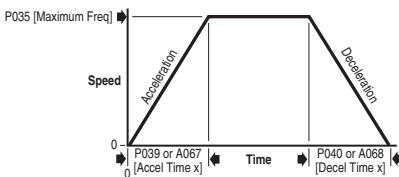
Options	0	"Drive Pot" (Default)	Internal frequency command from the potentiometer on the integral keypad.
	1	"InternalFreq"	Internal frequency command from A069 [Internal Freq].
	2	"0-10V Input"	External frequency command from the 0-10V analog input or remote potentiometer.
	3	"4-20mA Input"	External frequency command from the 4-20mA analog input.
	4	"Preset Freq"	External frequency command as defined by A070 - A073 [Preset Freq x] when A051 and A052 [Digital Inx Sel] are programmed as "Preset Frequencies" and the digital inputs are active.
	5	"Comm Port"	External frequency command from the communications port.

P039 [Accel Time 1] Related Parameter(s): [P038](#), [P040](#), [A051](#), [A052](#), [A067](#), [A070-A073](#)

Sets the rate of acceleration for all speed increases.

$$\frac{\text{Maximum Freq}}{\text{Accel Time}} = \text{Accel Rate}$$

Values	Default	10.0 Secs
	Min/Max:	0.0/600.0 Secs
	Display:	0.1 Secs



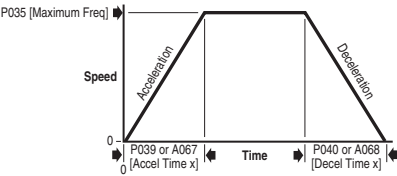
Basic Program Group *(continued)*

P040 [Decel Time 1]
Related Parameter(s): [P038](#), [P039](#), [A051](#), [A052](#), [A068](#), [A070-A073](#)

Sets the rate of deceleration for all speed decreases.

$$\frac{\text{Maximum Freq}}{\text{Decel Time}} = \text{Decel Rate}$$

Values	Default	10.0 Secs
	Min/Max:	0.1/600.0 Secs
	Display:	0.1 Secs



P041 [Reset To Defaults]



Stop drive before changing this parameter.

Resets all parameter values to factory defaults.

Options	0	"Idle State" (Default)	
	1	"Reset Defaults"	- After the reset function is complete, this parameter will set itself back to "0". - Causes an F48 Params Defaulted fault.

Advanced Program Group

A051 [Digital In1 Sel]

(I/O Terminal 5)

Related Parameter(s): [d012](#), [d014](#), [P038](#), [P039](#), [P040](#), [A067](#), [A068](#), [A070-A073](#), [A078](#), [A079](#)

A052 [Digital In2 Sel]

(I/O Terminal 6)



Stop drive before changing this parameter.

Selects the function for the digital inputs. Refer to the flowchart on [page 1-17](#) for more information on speed reference control priority.

Options	0 "Not Used"	Terminal has no function but can be read over network communications via d014 [Dig In Status]
	1 "Acc 2 & Dec 2"	- When active, A067 [Accel Time 2] and A068 [Decel Time 2] are used for all ramp rates except Jog. - Can only be tied to one input. Refer to the flowchart on page 1-18 for more information on Accel/Decel selection.
	2 "Jog"	- When input is present, drive accelerates according to the value set in A079 [Jog Accel/Decel] and ramps to the value set in A078 [Jog Frequency]. - When input is removed, drive ramps to a stop according to the value set in A079 [Jog Accel/Decel]. - A valid "Start" command will override this input.
	3 "Aux Fault"	When enabled, an F2 Auxiliary Input fault will occur when the input is removed.
	4 "Preset Freq" (Default)	Refer to A070 - A073 [Preset Freq x]. Important: Digital Inputs have priority for frequency control when programmed as a Preset Speed and are active. Refer to the flowchart on 1-17 for more information on speed reference control priority.
	5 "Local"	When active, sets integral keypad as start source and potentiometer on the integral keypad as speed source.
	6 "Comm Port"	- When active, sets communications device as default start/speed command source. - Can only be tied to one input.
	7 "Clear Fault"	When active, clears an active fault.
	8 "RampStop,CF"	Causes drive to immediately ramp to a stop regardless of how P037 [Stop Mode] is set.
	9 "CoastStop,CF"	Causes drive to immediately coast to a stop regardless of how P037 [Stop Mode] is set.
	10 "DCInjStop,CF"	Causes drive to immediately begin a DC Injection stop regardless of how P037 [Stop Mode] is set.
	11 "Jog Forward"	Drive accelerates to A078 [Jog Frequency] according to A079 [Jog Accel/Decel] and ramps to stop when input becomes inactive. A valid start will override this command.
	12 "Jog Reverse"	Drive accelerates to A078 [Jog Frequency] according to A079 [Jog Accel/Decel] and ramps to stop when input becomes inactive. A valid start will override this command.

A051 & A052 Options (Cont.)	13 “10V In Ctrl”	Selects 0-10V or $\pm 10V$ control as the frequency reference. Start source is not changed.
	14 “20mA In Ctrl”	Selects 4-20mA control as the frequency reference. Start source is not changed.
	15 - 25	Reserved
	26 “Anlg Invert”	Inverts the scaling of the analog input levels set in A110 [Anlg In 0-10V Lo] and A111 [Anlg In 0-10V Hi] or A112 [Anlg In4-20mA Lo] and A113 [Anlg In4-20mA Hi].

A055 [Relay Out Sel]Related Parameter(s): [P033](#), [A056](#), [A092](#)

Sets the condition that changes the state of the output relay contacts.

Options	0 “Ready/Fault” (Default)	Relay changes state when power is applied. This indicates that the drive is ready for operation. Relay returns drive to shelf state when power is removed or a fault occurs.
	1 “At Frequency”	Drive reaches commanded frequency.
	2 “MotorRunning”	Motor is receiving power from the drive.
	3 “Reverse”	Drive is commanded to run in reverse direction.
	4 “Motor Overld”	Motor overload condition exists.
	5 “Ramp Reg”	Ramp regulator is modifying the programmed accel/decel times to avoid an overcurrent or overvoltage fault from occurring.
	6 “Above Freq”	Drive exceeds the frequency (Hz) value set in A056 [Relay Out Level].
	7 “Above Cur”	Drive exceeds the current (% Amps) value set in A056 [Relay Out Level]. Important: Value for A056 [Relay Out Level] must be entered in percent of drive rated output current.
	8 “Above DCVolt”	Drive exceeds the DC bus voltage value set in A056 [Relay Out Level].
	9 “Retries Exst”	Value set in A092 [Auto Rstrt Tries] is exceeded.
	10 “Above Anlg V”	- Analog input voltage (I/O Terminal 13) exceeds the value set in A056 [Relay Out Level]. - This parameter setting can also be used to indicate a PTC trip point when the input (I/O Terminal 13) is wired to a PTC and external resistor. - Use A056 to set threshold.
	11-19	Reserved
	20 “ParamControl”	Enables the output to be controlled over network communications by writing to A056 [Relay Out Level]. (0 = Off, 1 = On.)
	21 “NonRec Fault”	Value set in A092 [Auto Rstrt Tries] is exceeded.

Advanced Program Group *(continued)*

A056 [Relay Out Level]

Related Parameter(s): [A055](#)



32 bit parameter.

Sets the trip point for the digital output relay if the value of [A055](#) [Relay Out Sel] is 6, 7, 8, 10 or 20.

A055 Setting	A056 Min/Max
6	0/240 Hz
7	0/180%
8	0/815 Volts
10	0/100%
20	0/1

Values	Default	0.0
	Min/Max:	0.0/9999
	Display:	0.1

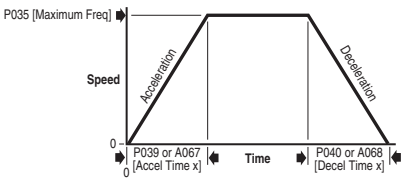
A067 [Accel Time 2]

Related Parameter(s): [P039](#)

When active, sets the rate of acceleration for all speed increases except jog. Refer to the flowchart on [page 1-18](#) for details.

$$\frac{\text{Maximum Freq}}{\text{Accel Time}} = \text{Accel Rate}$$

Values	Default	20.0 Secs
	Min/Max:	0.0/600.0 Secs
	Display:	0.1 Secs



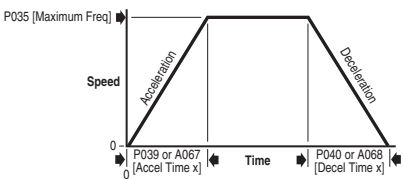
A068 [Decel Time 2]

Related Parameter(s): [P040](#)

When active, sets the rate of deceleration for all speed decreases except jog. Refer to the flowchart on [page 1-18](#) for details.

$$\frac{\text{Maximum Freq}}{\text{Decel Time}} = \text{Decel Rate}$$

Values	Default	20.0 Secs
	Min/Max:	0.1/600.0 Secs
	Display:	0.1 Secs



Advanced Program Group *(continued)*

A069 [Internal Freq]

Related Parameter(s): [P038](#)

Provides the frequency command to the drive when [P038](#) [Speed Reference] is set to 1 "Internal Freq". When enabled, this parameter will change the frequency command in "real time" using the integral keypad Up Arrow or Down Arrow when in program mode.

Important: Once the desired command frequency is reached, the Enter key must be pressed to store this value to EEPROM memory. If the ESC key is used before the Enter key, the frequency will return to the original value following the normal accel/decel curve.

Values	Default	60.0 Hz
	Min/Max:	0.0/240.0 Hz
	Display:	0.1 Hz

A070 [Preset Freq 0]⁽¹⁾

Related Parameter(s): [P038](#), [P039](#), [P040](#), [A051](#), [A052](#), [A067](#), [A068](#)

A071 [Preset Freq 1]

A072 [Preset Freq 2]

A073 [Preset Freq 3]

Values	A070 Default	0.0 Hz
	A071 Default	5.0 Hz
	A072 Default	10.0 Hz
	A073 Default	20.0 Hz
	Min/Max:	0.0/240.0 Hz
	Display:	0.1 Hz

Provides a fixed frequency command value when [A051](#) - [A052](#) [Digital Inx Sel] is set to 4 "Preset Frequencies".

An active preset input will override speed command as shown in the flowchart on [page 1-18](#).

⁽¹⁾ To activate A070 [Preset Freq 0] set [P038](#) [Speed Reference] to option 4 "Preset Freq 0-3".

Input State of Digital In 1 (I/O Terminal 05)	Input State of Digital In 2 (I/O Terminal 06)	Frequency Source	Accel / Decel Parameter Used ⁽²⁾
0	0	A070 [Preset Freq 0]	[Accel Time 1] / [Decel Time 1]
1	0	A071 [Preset Freq 1]	[Accel Time 1] / [Decel Time 1]
0	1	A072 [Preset Freq 2]	[Accel Time 2] / [Decel Time 2]
1	1	A073 [Preset Freq 3]	[Accel Time 2] / [Decel Time 2]

⁽²⁾ When a Digital Input is set to "Accel 2 & Decel 2", and the input is active, that input overrides the settings in this table.

A078 [Jog Frequency]

Related Parameter(s): [P035](#), [A051](#), [A052](#), [A079](#)

Sets the output frequency when a jog command is issued.

Values	Default	10.0 Hz
	Min/Max:	0.0/ P035 [Maximum Freq]
	Display:	0.1 Hz

Advanced Program Group *(continued)*

A079 [Jog Accel/Decel]

Related Parameter(s): [A051](#), [A052](#), [A078](#)

Sets the acceleration and deceleration time when a jog command is issued.

Values	Default	10.0 Secs
	Min/Max:	0.1/600.0 Secs
	Display:	0.1 Secs

A080 [DC Brake Time]

Related Parameter(s): [P037](#), [A081](#)

Sets the length of time that DC brake current is "injected" into the motor. Refer to parameter [A081](#) [DC Brake Level].

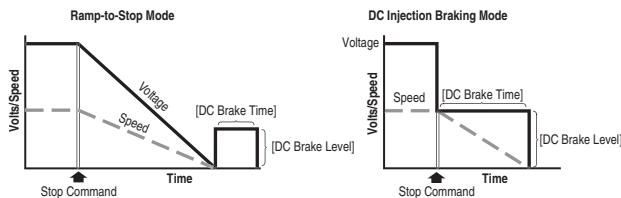
Values	Default	0.0 Secs
	Min/Max:	0.0/90.0 Secs
	Display:	0.1 Secs

A081 [DC Brake Level]

Related Parameter(s): [P037](#), [A080](#)

Defines the maximum DC brake current, in amps, applied to the motor when [P037](#) [Stop Mode] is set to either "Ramp" or "DC Brake".

Values	Default	Drive Rated Amps $\times$ 0.05
	Min/Max:	0.0/(Drive Rated Amps $\times$ 1.8)
	Display:	0.1 Amps



ATTENTION: If a hazard of injury due to movement of equipment or material exists, an auxiliary mechanical braking device must be used.



ATTENTION: This feature should not be used with synchronous or permanent magnet motors. Motors may be demagnetized during braking.

Advanced Program Group *(continued)*

A082 [DB Resistor Sel]



Stop drive before changing this parameter.

Enables/disables external dynamic braking.

Setting	Min/Max
0	"Disabled"
1	"Normal RA Res" (5% Duty Cycle) - Refer to Table B.C on page B-2
2	No Protection (100% Duty Cycle)
3-99	Duty Cycle Limited (3% – 99% of Duty Cycle)

Values	Default	0
	Min/Max:	0/99
	Display:	1

A083 [S Curve %]

Sets the percentage of acceleration or deceleration time that is applied to the ramp as S Curve. Time is added, 1/2 at the beginning and 1/2 at the end of the ramp.

Values	Default	0% (Disabled)
	Min/Max:	0/100%
	Display:	1%

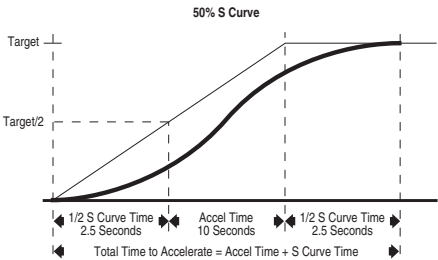
Example:

Accel Time = 10 Seconds

S Curve Setting = 50%

S Curve Time = $10 \times 0.5 = 5$ Seconds

Total Time = $10 + 5 = 15$ Seconds



Advanced Program Group *(continued)*

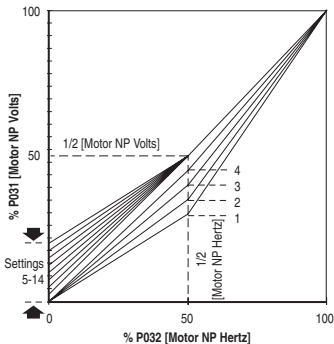
A084 [Boost Select]

Related Parameter(s): [d004](#), [P031](#), [P032](#)

Sets the boost voltage (% of [P031](#) [Motor NP Volts]) and redefines the Volts per Hz curve.⁽¹⁾

⁽¹⁾ Drive may add additional voltage unless option 5 is selected.

Options	1	"30.0, VT"	Variable Torque
	2	"35.0, VT"	
	3	"40.0, VT"	
	4	"45.0, VT"	
	5	"0.0 no IR"	
	6	"0.0"	Constant Torque
	7	"2.5, CT" [Default for 3.7kW (5HP) Drives]	
	8	"5.0, CT" (Default)	
	9	"7.5, CT"	
	10	"10.0, CT"	
	11	"12.5, CT"	
	12	"15.0, CT"	
	13	"17.5, CT"	
	14	"20.0, CT"	



A088 [Maximum Voltage]

Sets the highest voltage the drive will output.

Values	Default	Drive Rated Volts
	Min/Max:	20/Drive Rated Volts
	Display:	1 VAC

Advanced Program Group
(continued)

A089 [Current Limit]

Maximum output current allowed before current limiting occurs.

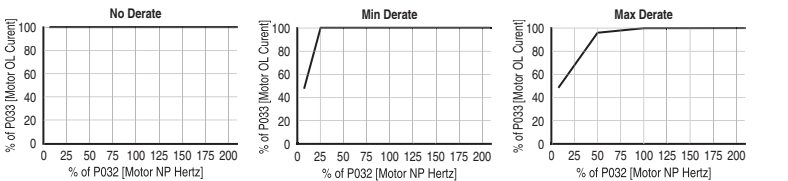
Values	Default	Drive Rated Amps × 1.5
	Min/Max:	0.1/(Drive Rated Amps × 1.8)
	Display:	0.1 Amps

A090 [Motor OL Select]

Related Parameter(s): [P032](#)

Drive provides Class 10 motor overload protection. Settings 0-2 select the derating factor for the I²t overload function.

Options	0	“No Derate” (Default)
	1	“Min Derate”
	2	“Max Derate”

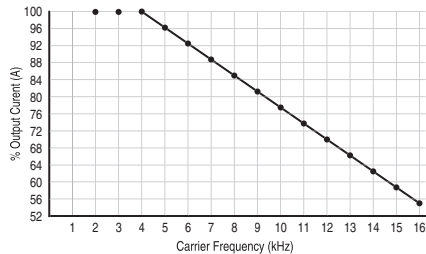


A091 [PWM Frequency]

Sets the carrier frequency for the PWM output waveform. The chart below provides derating guidelines based on the PWM frequency setting.

Important: Ignoring derating guidelines can cause reduced drive performance.

Values	Default	4.0 kHz
	Min/Max:	2.0/16.0 kHz
	Display:	0.1 kHz



Advanced Program Group *(continued)*

A092 [Auto Rstrt Tries]

Sets the maximum number of times the drive attempts to reset a fault and restart.

Clear a Type 1 fault and restart the drive.

1. Set A092 [Auto Rstrt Tries] to a value other than "0".
2. Set [A093](#) [Auto Rstrt Delay] to a value other than "0".

Clear an OverVoltage, UnderVoltage or Heatsink OvrTmp fault without restarting the drive.

1. Set A092 [Auto Rstrt Tries] to a value other than "0".
2. Set [A093](#) [Auto Rstrt Delay] to "0".



ATTENTION: Equipment damage and/or personal injury may result if this parameter is used in an inappropriate application. Do not use this function without considering applicable local, national and international codes, standards, regulations or industry guidelines.

Values	Default	0
	Min/Max:	0/9
	Display:	1

A093 [Auto Rstrt Delay]

Related Parameter(s): [A092](#)

Sets the time between restart attempts when [A092](#) [Auto Rstrt Tries] is set to a value other than zero.

Values	Default	1.0 Secs
	Min/Max:	0.0/300.0 Secs
	Display:	0.1 Secs

A094 [Start At PowerUp]



Stop drive before changing this parameter.

Enables/disables a feature that allows a Start or Run command to automatically cause the drive to resume running at commanded speed after drive input power is restored. Requires a digital input configured for Run or Start and a valid start contact.

This parameter will not function if parameter [P036](#) [Start Source] is set to 4 "2-W High Speed".



ATTENTION: Equipment damage and/or personal injury may result if this parameter is used in an inappropriate application. Do not use this function without considering applicable local, national and international codes, standards, regulations or industry guidelines.

Options	0	"Disabled" (Default)
	1	"Enabled"

Advanced Program Group *(continued)*

A095 [Reverse Disable]

Related Parameter(s): [d006](#)



Stop drive before changing this parameter.

Enables/disables the function that allows the direction of motor rotation to be changed. The reverse command may come from a digital command, the keypad or a serial command. All reverse inputs including two-wire Run Reverse will be ignored with reverse disabled.

Options	0	"Rev Enabled" (Default)
	1	"Rev Disabled"

A096 [Flying Start En]

Sets the condition that allows the drive to reconnect to a spinning motor at actual RPM.

Options	0	"Disabled" (Default)
	1	"Enabled"

A097 [Compensation]

Enables/disables correction options that may improve problems with motor instability.

Options	0	"Disabled"
	1	"Electrical" (Default) Some drive/motor combinations have inherent instabilities which are exhibited as non-sinusoidal motor currents. This setting attempts to correct this condition.
	2	"Mechanical" Some motor/load combinations have mechanical resonances which can be excited by the drive current regulator. This setting slows down the current regulator response and attempts to correct this condition.
	3	"Both"

A098 [SW Current Trip]

Related Parameter(s): [P033](#)

Enables/disables a software instantaneous (within 100 ms) current trip.

Values	Default	0.0 (Disabled)
	Min/Max:	0.0/(Drive Rated Amps × 2)
	Display:	0.1 Amps

A099 [Process Factor]

Related Parameter(s): [d010](#)

Scales the value displayed by [d010](#) [Process Display].

$$\begin{array}{c} \text{Output} \\ \text{Freq} \end{array} \times \begin{array}{c} \text{Process} \\ \text{Factor} \end{array} = \begin{array}{c} \text{Process} \\ \text{Display} \end{array}$$

Values	Default	30.0
	Min/Max:	0.1/999.9
	Display:	0.1

Advanced Program Group *(continued)*

A100 [Fault Clear]



Stop drive before changing this parameter.

Resets a fault and clears the fault queue. Used primarily to clear a fault over network communications.

Options 0 "Ready/Idle" (Default)

1 "Reset Fault"

2 "Clear Buffer" (Parameters [d007-d009](#) [Fault x Code])

A101 [Program Lock]

Protects parameters against change by unauthorized personnel.

Options 0 "Unlocked" (Default)

1 "Locked"

A102 [Testpoint Sel]

Used by Rockwell Automation field service personnel.

Values Default 400

Min/Max: 0/FFFF

Display: 1 Hex

A103 [Comm Data Rate]

Related Parameter(s): [d015](#)

Sets the serial port rate for the RS485 (DSI) port.

Important: Power to drive must be cycled before any changes will affect drive operation.

Options 0 "1200"

1 "2400"

2 "4800"

3 "9600" (Default)

4 "19.2K"

5 "38.4K"

A104 [Comm Node Addr]

Related Parameter(s): [d015](#)

Sets the drive node address for the RS485 (DSI) port if using a network connection.

Important: Power to drive must be cycled before any changes will affect drive operation.

Values Default 100

Min/Max: 1/247

Display: 1

Advanced Program Group *(continued)*

A105 [Comm Loss Action]

Related Parameter(s): [d015](#), [P037](#), [A106](#)

Selects the drive's response to a loss of the communication connection or excessive communication errors.

Options	0	"Fault" (Default)	Drive will fault on an F81 Comm Loss and coast to stop.
	1	"Coast to Stop"	Stops drive via coast to stop.
	2	"Stop"	Stops drive via P037 [Stop Mode] setting.
	3	"Continu Last"	Drive continues operating at communication commanded speed saved in RAM.

A106 [Comm Loss Time]

Related Parameter(s): [d015](#), [A105](#)

Sets the time that the drive will remain in communication loss before implementing the option selected in [A105](#) [Comm Loss Action].

Values	Default	5.0 Secs
	Min/Max:	0.1/60.0 Secs
	Display:	0.1 Secs

A107 [Comm Format]

Selects the protocol (RTU only), data bits (8 data bits only), parity (None, Even, Odd), and stop bits (1 stop bit only) used by the RS485 port on the drive.

Refer to [Appendix C](#) for details on using the drive communication features.

Important: Power to drive must be cycled before any changes will affect drive operation.

Options	0	"RTU 8-N-1" (Default)
	1	"RTU 8-E-1"
	2	"RTU 8-O-1"
	3	"RTU 8-N-2"
	4	"RTU 8-E-2"
	5	"RTU 8-O-2"

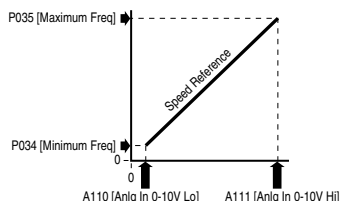
A110 [Anlg In 0-10V Lo]Related Parameter(s): [d020](#), [P034](#), [P038](#), [A051](#), [A052](#)

Stop drive before changing this parameter.

Sets the analog input level that corresponds to P034 [Minimum Freq] if a 0-10V input is used by P038 [Speed Reference].

Analog inversion can be accomplished by setting this value larger than A111 [Anlg In 0-10V Hi] or by setting [A051](#) - [A052](#) [Digital Inx Sel] to option 26 "Anlg Invert".

Values	Default	0.0%
	Min/Max:	0.0/100.0%
	Display:	0.1%

**A111 [Anlg In 0-10V Hi]**Related Parameter(s): [d020](#), [P035](#), [P038](#), [A051](#), [A052](#)

Sets the analog input level that corresponds to P035 [Maximum Freq] if a 0-10V input is used by P038 [Speed Reference].

Analog inversion can be accomplished by setting this value smaller than A110 [Anlg 0-10V In Lo] or by setting [A051](#) - [A052](#) [Digital Inx Sel] to option 26 "Anlg Invert".

Values	Default	100.0%
	Min/Max:	0.0/100.0%
	Display:	0.1%

A112 [Anlg In4-20mA Lo]Related Parameter(s): [d021](#), [P034](#), [P038](#), [A051](#), [A052](#)

Sets the analog input level that corresponds to P034 [Minimum Freq] if a 4-20mA input is used by P038 [Speed Reference].

Analog inversion can be accomplished by setting this value larger than A113 [Anlg In4-20mA Hi] or by setting [A051](#) - [A052](#) [Digital Inx Sel] to option 26 "Anlg Invert".

Values	Default	0.0%
	Min/Max:	0.0/100.0%
	Display:	0.1%

A113 [Anlg In4-20mA Hi]Related Parameter(s): [d021](#), [P035](#), [P038](#), [A051](#), [A052](#)

Sets the analog input level that corresponds to P035 [Maximum Freq] if a 4-20mA input is used by P038 [Speed Reference].

Analog inversion can be accomplished by setting this value smaller than A112 [Anlg In4-20mA Lo] or by setting [A051](#) - [A052](#) [Digital Inx Sel] to option 26 "Anlg Invert".

Values	Default	100.0%
	Min/Max:	0.0/100.0%
	Display:	0.1%

A114 [Slip Hertz @ FLA]

Related Parameter(s): [P033](#)

Compensates for the inherent slip in an induction motor. This frequency is added to the commanded output frequency based on motor current.

Values	Default	2.0 Hz
	Min/Max:	0.0/10.0 Hz
	Display:	0.1 Hz

A115 [Process Time Lo]

Related Parameter(s): [d010](#), [P034](#)

Scales the time value when the drive is running at [P034](#) [Minimum Freq]. When set to a value other than zero, [d010](#) [Process Display] indicates the duration of the process.

Values	Default	0.00
	Min/Max:	0.00/99.99
	Display:	0.01

A116 [Process Time Hi]

Related Parameter(s): [d010](#), [P035](#)

Scales the time value when the drive is running at [P035](#) [Maximum Freq]. When set to a value other than zero, [d010](#) [Process Display] indicates the duration of the process.

Values	Default	0.00
	Min/Max:	0.00/99.99
	Display:	0.01

Parameter Cross Reference – by Name

<u>Parameter Name</u>	<u>No.</u>	<u>Group</u>	<u>Parameter Name</u>	<u>No.</u>	<u>Group</u>
Accel Time 1	P039	Basic Program	Fault Clear	A100	Advanced Program
Accel Time 2	A067	Advanced Program	Flying Start En	A096	Advanced Program
Analog In 0-10V	d020	Display	Internal Freq	A069	Advanced Program
Analog In 4-20mA	d021	Display	Jog Accel/Decel	A079	Advanced Program
Anlg In 0-10V Hi	A111	Advanced Program	Jog Frequency	A078	Advanced Program
Anlg In 0-10V Lo	A110	Advanced Program	Maximum Freq	P035	Basic Program
Anlg In4-20mA Hi	A113	Advanced Program	Maximum Voltage	A088	Advanced Program
Anlg In4-20mA Lo	A112	Advanced Program	Minimum Freq	P034	Basic Program
Auto Rstrt Delay	A093	Advanced Program	Motor NP Hertz	P032	Basic Program
Auto Rstrt Tries	A092	Advanced Program	Motor NP Volts	P031	Basic Program
Boost Select	A084	Advanced Program	Motor OL Current	P033	Basic Program
Comm Data Rate	A103	Advanced Program	Motor OL Select	A090	Advanced Program
Comm Format	A107	Advanced Program	Output Current	d003	Display
Comm Loss Action	A105	Advanced Program	Output Freq	d001	Display
Comm Loss Time	A106	Advanced Program	Output Voltage	d004	Display
Comm Node Addr	A104	Advanced Program	Preset Freq x	A070-A073	Advanced Program
Comm Status	d015	Display	Process Display	d010	Display
Commanded Freq	d002	Display	Process Factor	A099	Advanced Program
Compensation	A097	Advanced Program	Process Time Hi	A116	Advanced Program
Contrl In Status	d013	Display	Process Time Lo	A115	Advanced Program
Control Source	d012	Display	Program Lock	A101	Advanced Program
Control SW Ver	d016	Display	PWM Frequency	A091	Advanced Program
Current Limit	A089	Advanced Program	Relay Out Level	A056	Advanced Program
DB Resistor Sel	A082	Advanced Program	Relay Out Sel	A055	Advanced Program
DC Brake Level	A081	Advanced Program	Reset To Defaults	P041	Basic Program
DC Brake Time	A080	Advanced Program	Reverse Disable	A095	Advanced Program
DC Bus Voltage	d005	Display	S Curve %	A083	Advanced Program
Decel Time 1	P040	Basic Program	Slip Compensation	A114	Advanced Program
Decel Time 2	A068	Advanced Program	Speed Reference	P038	Basic Program
Dig In Status	d014	Display	Start At PowerUp	A094	Advanced Program
Digital Inx Sel	A051, A052	Advanced Program	Start Source	P036	Basic Program
Drive Status	d006	Display	Stop Mode	P037	Basic Program
Drive Temp	d024	Display	SW Current Trip	A098	Advanced Program
Drive Type	d017	Display	Testpoint Data	d019	Display
Elapsed Run Time	d018	Display	Testpoint Sel	A102	Advanced Program
Fault x Code	d007-d009	Display			

Notes: